

# Dividend Risk in an Autocallable Reverse Convertible

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## 1 Project Overview

The objective of this project is to price an **Autocallable Reverse Convertible** product using a Local Volatility (LV) framework and to evaluate the impact of dividend risk.

The workflow is divided into three main phases:

1. **Calibration:** Implement a semi-parametric local volatility model that accounts for discrete proportional dividends using a fixed-point algorithm.
2. **Pricing:** Value specific vanilla products and the autocallable structured product via Monte Carlo simulation.
3. **Risk Management:** Construct a hedging portfolio (Delta, Vega, and Dividend hedge) and perform a P&L analysis to assess the residual risks of the structure.

## 2 Market Data

You are provided with market data for an underlying equity: **ISP IM**. For each trading day, the dataset includes:

- The **mid-implied volatility smile**  $IV_t(T, M)$  for a certain time-to-maturity  $T$  (Act/365) and moneyness  $M = K/S_t$ .
- A strip of **SSDF** (Single Stock Dividend Future) prices  $SSDF_t(T)$  for the years 2026, 2027, and 2028. Assume full dividend payments on: 18-May-2026, 24-May-2027, and 22-May-2028.
- Rates (discounting/funding/repo) and dividend tax rates are assumed to be **zero** for simplicity.

## 3 Local Volatility Framework

Following the approach of [1], you must calibrate the LV model for each underlying.

### 3.1 Functional Form

Assume a constant in time local volatility  $\sigma_{LV}(S)$ . We use a semi-parametric squared local volatility with linear extrapolation in log-strike:

$$\sigma_{LV}^2(S) = \begin{cases} a + b \ln(S) & S < S_{low} \\ f(\ln(S)) & S \in [S_{low}, S_{high}] \\ c + d \ln(S) & S > S_{high} \end{cases} \quad (1)$$

$f(X)$  can be built as a linear interpolation function on certain nodes  $(X_j, \sigma_j^2)$  between  $\ln(S_{low})$  and  $\ln(S_{high})$ .  $S_{low}$  and  $S_{high}$  should be set at  $\pm 3$  standard deviations from the forward price.

### 3.2 Discrete Dividends

Dividends are modeled as proportional to the spot price at the ex-date  $\tau_i$ :

$$S_{\tau_i^+} = S_{\tau_i^-}(1 - d_i) \quad (2)$$

The coefficient  $d_i$  must be implied from the Dividend Future prices to ensure the model is consistent with market prices.

### 3.3 The Dual PDE (Fokker-Planck)

To obtain European Call prices  $C(K, T)$  for all strikes and maturities, solve the Forward Kolmogorov Equation (Dual PDE). For a local volatility  $\sigma(K, T)$ , the equation is:

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma^2(K)K^2\frac{\partial^2 C}{\partial K^2} - (r - q)K\frac{\partial C}{\partial K} - qC \quad (3)$$

$$C(K, 0) = (S_0 - K)^+ \quad (4)$$

Across every dividend date we have:

$$\mathbb{E}[(S_{\tau_i^+} - K)^+] = \mathbb{E}[(S_{\tau_i^-}(1 - d_i) - K)^+] \quad (5)$$

so that the following continuity condition holds:

$$C(K, \tau_i^+) = (1 - d_i)C\left(\frac{K}{1 - d_i}, \tau_i^-\right) \quad (6)$$

**Implementation Note:** Use an implicit finite difference scheme. Pay particular attention to the continuity condition at each dividend date  $\tau_i$  and the restart of the PDE at each maturity  $T_i$  with the appropriate local volatility  $\sigma_{LV,i}$ . Adopting a change of variable as suggested in [2] can be beneficial for achieving better precision and simpler boundary conditions. Look in the appendix for further details.

## 4 Calibration via Fixed-Point Algorithm

The calibration of  $\sigma_{LV}$  is an inverse problem. Use a bootstrap procedure by means of a fixed-point iteration:

1. Initialize  $\sigma_{LV}^{(0)}(S)$  (e.g., using the market implied volatility).
2. Solve the Dual PDE to obtain model prices  $C_{mod}$  and compute model implied volatilities  $IV_{mod}^{(k)}$ .
3. Update the local volatility using the fixed-point algorithm in [1].
4. Iterate until the error  $\max |IV_{mkt} - IV_{mod}^{(k)}| < 0.01\%$  to obtain the calibrated local volatility  $\sigma_{LV}(S)$ .

## 5 Monte Carlo Pricing

Use a Monte Carlo simulation to price:

- Vanilla Put At-The-Money Forward (ATMF) options with 1Y, 2Y, and 3Y expiries.
- Vanilla Put options with an 80% strike and 1Y, 2Y, and 3Y expiries.

- An Autocallable Reverse Convertible certificate as described in the attached termsheet.

Compare results from both the Black-Scholes model and the Local Volatility model to double-check the calibration quality. Explain any differences observed between the two models regarding the autocallable product. Further details on the Monte Carlo simulation of the Local Volatility model can be found in the appendix.

## 5.1 P&L Explain and Backtesting

Construct a portfolio with a short position of 20M EUR in the certificate. Perform the following hedges at  $t_0$ :

- **Portfolio A:** Delta hedge only.
- **Portfolio B:** Delta + Dividend hedge (using SSDFs).
- **Portfolio C:** Delta + Dividend + Vega hedge (using an ATMF vanilla option, 3Y expiry).
- **Portfolio D:** Delta + Dividend + Optimized Vega hedge (selecting the optimal vanilla instrument as identified in the analysis below).

For each of the four portfolios above:

1. Plot the Vega and Dividend Sensitivity against spot prices in the range  $[0.4S_{t_0}, 1.2S_{t_0}]$ .
2. Identify the best choice of a vanilla option (in terms of strike and maturity) for Portfolio D such that, in your opinion, the risks calculated above are best mitigated.
3. Evaluate the P&L on dates  $t_1$  and  $t_2$ .
4. Discuss the impact of **Dividend Risk**, especially considering how the autocallable feature amplifies this sensitivity compared to a vanilla product.
5. Assess if the delta, dividend, and vega hedges are sufficient to mitigate relevant financial losses. Discuss any remaining unhedged risks.
6. Assuming a Bid-Offer spread of 0.5% for the reverse convertible, 0.2% for the vanilla options, 0.1% for SSDFs, and 0.01% for the spot price, discuss the profitability of selling the certificate and hedging with the aforementioned instruments.
7. Suggest any alternative strategies that could improve the performance of the trading desk.

**Attention:** In calculating the hedge ratios for the four portfolios, ensure consistency in the Greeks calculation for each product. Specifically, use the same model for the Greeks of both the certificate and the vanilla options, and evaluate whether there is any Vega or Delta exposure arising from the dividend futures.

## A Implicit Finite Difference Method

The Finite Difference Method is probably the simplest tool for solving partial differential equations, in particular when there is only one state variable. Considering the equation:

$$\partial_t P(t, x) = r(t, x)P(t, x) + c(t, x)\partial_x P(t, x) + d(t, x)\partial_{xx} P(t, x) \quad (7)$$

### A.1 Time Discretization

The first step is to discretize the time derivative:

$$\partial_t P(t, x) = \frac{P(t + \Delta t, x) - P(t, x)}{\Delta t} + O(\Delta t) \quad (8)$$

writing the PDE as:

$$\partial_t P(t, x) = \mathcal{L}(t, x)P(t, x) \quad (9)$$

with:

$$\mathcal{L}(t, x)P(t, x) = r(t, x)P(t, x) + c(t, x)\partial_x P(t, x) + d(t, x)\partial_{xx} P(t, x) \quad (10)$$

the Implicit Finite Difference Scheme is given by:

$$\frac{P(t + \Delta t, x) - P(t, x)}{\Delta t} = \mathcal{L}(t + \Delta t, x)P(t + \Delta t, x) \quad (11)$$

or

$$(1 - \mathcal{L}(t + \Delta t, x)\Delta t)P(t + \Delta t, x) = P(t, x) \quad (12)$$

### A.2 Space Discretization

The space discretization is done through centered finite differences. Consider a uniform grid of points  $x_i$ ,  $i = 0, \dots, N_x - 1$ , then for the first derivative we have:

$$\partial_x P(t, x_i) = \frac{P(t, x_{i+1}) - P(t, x_{i-1})}{2\Delta x} + O(\Delta x^2) \quad (13)$$

the second derivative is given by:

$$\partial_{xx} P(t, x) = \frac{P(t, x_{i+1}) - 2P(t, x_i) + P(t, x_{i-1}))}{\Delta x^2} + O(\Delta x^2) \quad (14)$$

so that we can discretize  $\mathcal{L}$  as:

$$\begin{aligned} L(t, x_i)P(t, x_i) &= r(t, x_i)P(t, x_i) \\ &+ c(t, x_i)\frac{P(t, x_{i+1}) - P(t, x_{i-1}))}{2\Delta x} \\ &+ d(t, x_i)\frac{P(t, x_{i+1}) - 2P(t, x_i) + P(t, x_{i-1}))}{\Delta x^2} \end{aligned} \quad (15)$$

Together with the implicit time discretization we obtain:

$$(1 - \Delta t L(t_{j+1}, x_i))P(t_{j+1}, x_i) = P(t_j, x_i) \quad (16)$$

With a slight change of notation  $P_{i,j} = P(t_j, x_i)$  we can write this extended form:

$$\begin{aligned} &\left(1 - \Delta t r(t_{j+1}, x_i) + 2\frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i)\right) P_{i,j+1} + \\ &\left(-\frac{\Delta t}{2\Delta x} c(t_{j+1}, x_i) - \frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i)\right) P_{i+1,j+1} + \\ &\left(+\frac{\Delta t}{2\Delta x} c(t_{j+1}, x_i) - \frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i)\right) P_{i-1,j+1} = P_{i,j} \end{aligned} \quad (17)$$

In matrix form this becomes:

$$A P_{j+1} = P_j \quad (18)$$

$$P_{j+1} = A^{-1} P_j \quad (19)$$

Notice that A is triadiagonal:

$$A_{i,i} = 1 - \Delta t r(t_{j+1}, x_i) + 2 \frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i) \quad (20)$$

$$A_{i,i+1} = -\frac{\Delta t}{2\Delta x} c(t_{j+1}, x_i) - \frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i) \quad (21)$$

$$A_{i,i-1} = +\frac{\Delta t}{2\Delta x} c(t_{j+1}, x_i) - \frac{\Delta t}{\Delta x^2} d(t_{j+1}, x_i) \quad (22)$$

and zero everywhere else. A is squared with  $N_x$  rows and columns.

### A.3 Boundary Conditions

We have obtained an iterative process to calculate the solution at a certain date. At the first time instant we need to define:

$$P(0, x) = \Phi(x) \quad (23)$$

for a certain function  $\Phi$ , so that we can use

$$P_{i,0} = \Phi(x_i) \quad (24)$$

as our starting point. Notice that the equations seen in the preceding section are well defined only inside the domain. Sometimes it is possible to have simple boundary condition (Dirichlet) of the form:

$$P(t, x_0) = lb(t) \quad (25)$$

$$P(t, x_{N_x-1}) = rb(t) \quad (26)$$

so that A takes the form:

$$A_{0,0} = 1 \quad (27)$$

$$A_{N_x-1, N_x-1} = 1 \quad (28)$$

but you may need to override the value on the previous iteration:

$$P_{j+1} = A^{-1} \hat{P}_j \quad (29)$$

$$\hat{P}_{0,j} = lb(t_j) \quad (30)$$

$$\hat{P}_{N_x-1,j} = rb(t_j) \quad (31)$$

### A.4 Change of Variable

It is suggested to use a change of variable as in [2] of the form:

$$y = \frac{S}{S + S_0} \in [0, 1] \quad (32)$$

$$S = S_0 \frac{y}{1 - y} \quad (33)$$

for which a new equation is obtained. The boundaries of the space domain can be taken as:

$$y_{min} = 0 \quad (34)$$

$$S_{min} = 0 \quad (35)$$

$$y_{max} = 1 - \frac{1}{N_y} \quad (36)$$

$$S_{max} = S_0(N_y - 1) \quad (37)$$

## A.5 Continuity Conditions

In many cases we need to define a continuity condition at certain particular dates where the solution of the equation is changing in a finite amount of time. The equation could take the form:

$$P(\hat{t}^+, x) = a(\hat{t}^-, x, P(\hat{t}^-, b(\hat{t}^-, x))) \quad (38)$$

these situations are managed in the following way:

1. Calculate the solution  $P(t_j, x_i)$  until  $t_j = \hat{t}$ . This will be your solution in  $\hat{t}^-$ .
2. For every  $x_i$  evaluate  $a_i = a(\hat{t}^-, x_i, P(\hat{t}^-, b(\hat{t}^-, x_i)))$ . You may need to interpolate to get the solution in  $b(\hat{t}^-, x_i)$ .
3. Update the solution  $P(t_j^+, x_i) = a_i$ .
4. Restart the equation in  $t_j$  from this new initial condition.

## A.6 Final Remarks

The equation we need to solve for the calibration of the local volatility model (with zero rates and repos) is given by:

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma^2(K)K^2\frac{\partial^2 C}{\partial K^2} \quad (39)$$

$$C(K, \tau_i^+) = (1 - d_i)C\left(\frac{K}{1 - d_i}, \tau_i^-\right) \quad (40)$$

$$C(K, 0) = (S_0 - K)^+ \quad (41)$$

Notice that with the suggested change of variable, the equation in  $y = 0$  and  $y = 1$  takes a particular simple form, so that the equation can be solved also on the boundaries. Given that the equation coefficients are constant in time, the matrix  $A$  can be computed and eventually factorized once, and used for every iteration until maturity.

## B Monte Carlo Simulation

The local volatility SDE with constant in time volatility is given by:

$$dS_t = \sigma(S_t)S_t dW_t \quad (42)$$

taking the logarithm we get  $X_t = \log(S_t)$ :

$$dX_t = \sigma(e^{X_t})dW_t - \frac{1}{2}\sigma^2(e^{X_t})dt \quad (43)$$

from which we can get a simple Euler Scheme:

$$X_{t+\Delta t} = X_t + \sigma(e^{X_t})\sqrt{\Delta t}N(0, 1) - \frac{1}{2}\sigma^2(e^{X_t})\Delta t \quad (44)$$

Getting back to  $S_t$  we obtain:

$$\begin{cases} S_{t+\Delta t} &= S_t e^{\sigma(S_t)\sqrt{\Delta t}N(0,1) - \frac{1}{2}\sigma^2(S_t)\Delta t} \\ S_{\tau_i^+} &= S_{\tau_i^-}(1 - d_i) \end{cases} \quad (45)$$

Smaller values of  $\Delta t$  will give a better precision but will increase computational time. A value of  $\Delta t = 2/365$  is usually enough to get the needed precision, but be sure to have a time grid that passes through every dividend date  $\tau$ .

## References

- [1] Reghai, A., Boya, G., & Vong, G. (2012). *Local Volatility Calibration with Discrete Dividends*. SSRN 2008215.
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